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24/03/2026.

Assignment :

1] Simple Linear Regression
→ It models the relationship b/w one independent variable & one dependent variable using a straight line.

2] Multiple Linear Regression.
→ It models the relationship b/w one independent variable and multiple independent variables.

3] Mathematical implementation of Simple Linear Regression.
→ Equation : $y = a_0 + a_1 x + \varepsilon$

4] Mathematical implementation of Multiple Linear Regression.
→ Equation : $Y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$.

5] Loss Function.
→ It measures the errors between predicted & actual values.

6] Polynomial Regression.
→ It models non-linear relationships using polynomial equation.

7] Optimization
→ It is the process of finding the best model parameters to ~~minize~~ minimize error.

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8] Evaluation Metrics

→ ✓ Mean Squared Error (MSE)
✓ R-squared
✓ Mean Absolute Error (MAE)
✓ Root Mean Square Error (RMSE)

9] Gradient Descent

→ An optimization algorithm used to minimize the cost function by updating parameters iteratively.

10] Assumption of Linear Regression.

→ ✓ Linear relationship.
✓ No multicollinearity.
✓ Homoscedasticity
✓ Normal distribution of errors.

11] Overfitting

→ Occurs when a model performs well on training data but poorly on test data.

12] Underfitting

→ Occurs when a model fails to capture patterns in training data.

13] Degree in Polynomial Regression

→ The degree is choosen based on ~~top~~ data complexity to bet fit non-linear patterns.

14] Mean Squared Error (MSE).

→ Average of squareed differences between predicted & actual values.

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15] Mean Absolute Error (MAE).

→ Average of absolute differences b/w predicted & actual values.

16] Root Mean Squared Error (RMSE)

→ Square root of MSE, indicating predicting error magnitude.

17] Lasso Regression

→ L1 regularization reduces model complexity & can shrink cofficients to zero, useful for fatures selection.

18] Ridge Regression

→ L2 regularization reduces overfitting by shrking coefficients but not to zero. It improves model stability.

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